

EE445 Mod2-Lec3: Least Squares Regression

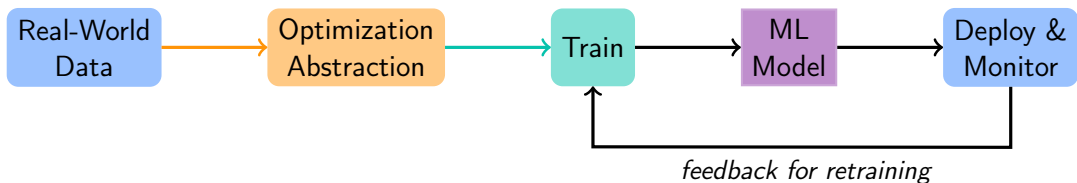
References:

- [VMLS]: Chapter 12

Agenda

- Optimality via linear algebra (eigenvalues of Hessian & linear independence)
-

ML workflow: Role of Linear Algebra & Optimization



This course so far:

- **Optimization & Linear Algebra:** Data $\rightarrow$ Optimization Abstraction via losses as norms
- **ML models:** Distance metric based models: k-NN, k-means

Coming up next:

- **Training:** Optimizing "convex" objectives (least squares)
- **ML models:** Linear models: linear regression, PCA, attention

ML workflow: Optimization Framework for Learning

Real world task involving data $\longrightarrow$ Optimization Problem $\longrightarrow$ ML model

Canonical Optimization Problem

$$\min_{\theta \in \Theta \subseteq \mathbb{R}^n} \ell(\theta)$$

Terminology:

- θ : Parameter/optimization or decision variable
- $\ell(\theta)$: Objective/Loss function; in ML will often depend on data Z ; write as $\ell(\theta; Z)$.
- $\Theta \subseteq \mathbb{R}^n$: Constraint set

Linear Models: Predicting Outputs from Inputs

- **Goal:** learn a function (aka "model") $f_{\theta}(x)$ that predicts quantity $y \in \mathbb{R}$ from input $x \in \mathbb{R}^d$.
- For linear models:

$$f_{\theta}(x) = \theta^{\top} x = \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d \quad (\text{linear})$$

$$f_{\theta}(x) = \theta^{\top} \begin{bmatrix} x \\ 1 \end{bmatrix} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d \quad (\text{affine})$$

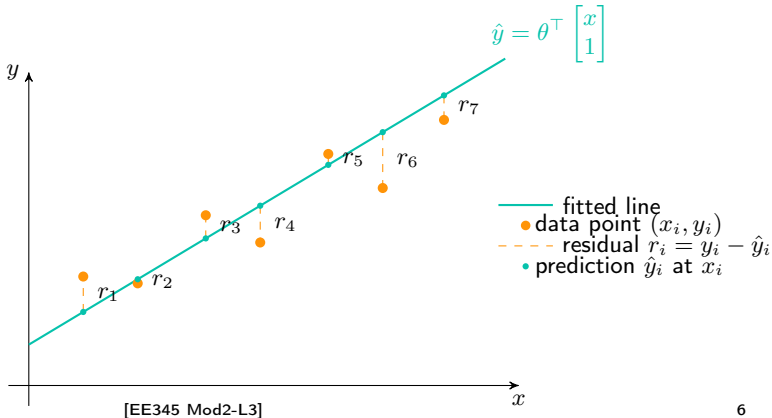
- Given data (x_i, y_i) for $i = 1, \dots, n$, we choose θ to minimize prediction error.
- Examples:
 - ▶ Predict house price from size, bedrooms, age.
 - ▶ Predict exam score from study hours.

Measuring Predictor Error in 2D (Linear Regression)

For a fitted line $\hat{y} = \theta_1 x + \theta_0$ and points (x_i, y_i) , the **residual** is

$$r_i = y_i - \hat{y}_i = y_i - \theta x_i - \theta_0, \quad \text{and} \quad \mathcal{L}(\theta_1, \theta_0) = \sum_{i=1}^n r_i^2.$$

OLS loss: minimize total squared residual length $\sum_i \|r_i\|_2^2$ (vertical errors).



Empirical Risk Minimization (ERM): Quintentessential ML Problem

Goal of supervised learning: Find a function that makes good predictions on new data.

Challenge: We don't know the true data distribution; indeed, we only see a **finite** set of samples.

Idea of ERM:

$$\min_{\theta} \frac{1}{n} \sum_{i=1}^n \ell(f_{\theta}(x_i), y_i) \quad \text{average over } n \text{ samples} = \text{empirical risk}$$

- f_{θ} : model with parameters θ
- $\ell(\cdot, \cdot)$: loss measuring prediction error (recall, norms and distances)

Why it matters: ERM formalizes the idea of *fitting a model to data*.

- Almost every ML algorithm—from linear regression to deep networks—minimizes some form of empirical risk.
- Least squares is the simplest example.

Least Squares (Linear Regression) Set-up

- Consider an $n \times d$ matrix $X \in \mathbb{R}^{n \times d}$ and vectors $y \in \mathbb{R}^n$, $\theta \in \mathbb{R}^d$

Least Squares Optimization Problem

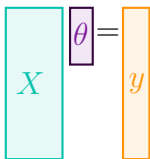
Find θ such that $X\theta = y$ that minimizes the ℓ_2 loss:

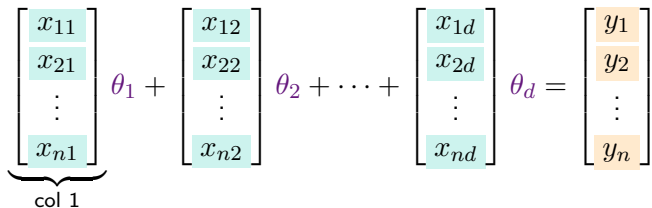
$$\min_{\theta \in \mathbb{R}^d} \mathcal{L}(\theta) = \min_{\theta \in \mathbb{R}^d} \|y - X\theta\|_2^2 = \min_{\theta \in \mathbb{R}^d} \sum_{i=1}^n \underbrace{(y_i - x_i^\top \theta)^2}_{\ell(f_\theta(x_i), y_i)}$$

- X is a matrix of training data; e.g., n is number of samples, d is number of 'features'
- $y \in \mathbb{R}^n$ contains 'target values', 'labels', or observations of real world phenomena
- $\theta \in \mathbb{R}^d$ is a set of feature weights

Overdetermined System of Equations $n \gg d$

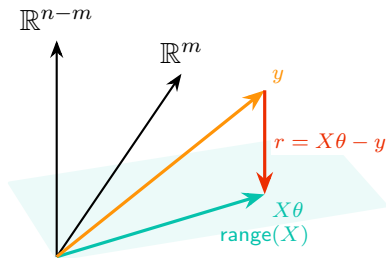
- **Goal:** Find a solution to $X\theta = y$ —that is, find θ such that $X\theta = y$
- **Overdetermined System:** X is a 'tall' matrix $\implies$ there are more samples (n) than features to choose (d).


$$X\theta = y$$


$$\begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{bmatrix} \theta_1 + \begin{bmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{bmatrix} \theta_2 + \cdots + \begin{bmatrix} x_{1d} \\ x_{2d} \\ \vdots \\ x_{nd} \end{bmatrix} \theta_d = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

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Projection Geometry: $y = X\theta + r$ with $r \perp \text{range}(X)$



$$m = \dim(\text{range}(X))$$

- $\text{range}(X)$ is the span of the columns of X :

$$\text{range}(X) = \{y' \in \mathbb{R}^n \mid \exists \theta' \in \mathbb{R}^d, y' = X\theta'\}$$

- $\text{range}(X)$ is also called the column space of X

Finding Minima and Convexity

Goal: Find θ^* that minimizes a differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$.

Conditions for Minima

first order condition: $\nabla f(\theta^*) = 0$ (stationary point)

Second-order condition : $v^\top \underbrace{\nabla^2 f(\theta^*)}_{=: H(\theta^*)} v > 0$ for all $v \neq 0$, (convexity)

- Gradient = direction of steepest ascent.
- Setting it to zero $\implies$ flat point.
- Positive-definite Hessian $\implies$ surface curves upward in all directions.
- $H(\theta) = \nabla^2 f(\theta)$ is the Hessian matrix.
- we say f is **strictly convex** near θ^*
- θ^* is the unique minimum.

Takeaway: Convex $\implies$ any stationary point is a global minimum.

Finding Minima in One Dimension

For a smooth function $f : \mathbb{R} \rightarrow \mathbb{R}$, a minimum occurs when:

$$f'(x^*) = 0 \quad \text{and} \quad f''(x^*) > 0.$$

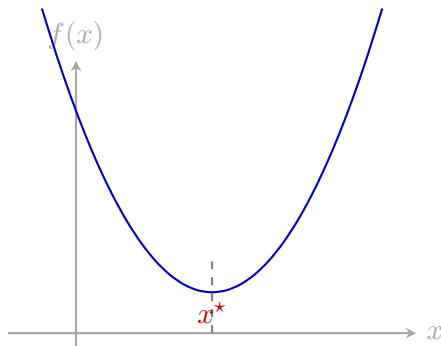
- $f'(x)$ measures slope: setting it to 0 finds flat points.
- $f''(x)$ measures curvature: if it's positive, the function curves upward (a bowl).

Example: $f(x) = (x - 2)^2 + 1$

$$f'(x) = 2(x - 2), \quad f''(x) = 2 > 0.$$

Minimum at $x^* = 2$.

Takeaway: In 1D, convexity means *positive curvature* — a bowl-shaped graph.



Minima in Two Dimensions

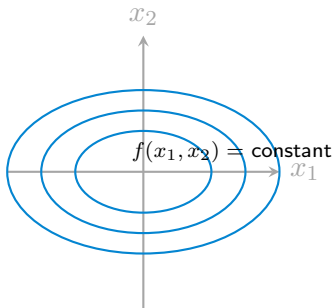
Now $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ has a gradient and Hessian:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}, \quad H = \nabla^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}.$$

Example: $f(x_1, x_2) = ax_1^2 + bx_2^2$ for $a, b > 0$.

$$H = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \implies v^\top H v = av_1^2 + bv_2^2.$$

$\implies H \succ 0$ (i.e. positive definite) if $a > 0$ and $b > 0$.



A More General 2D Hessian: Tilted Ellipses

Example (coupled quadratic):

$$f(x_1, x_2) = \frac{1}{2} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \underbrace{\begin{bmatrix} a & c \\ c & b \end{bmatrix}}_H \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (a, b > 0).$$

Gradient & Hessian:

$$\nabla f(x) = H x, \quad \nabla^2 f(x) = H = \begin{bmatrix} a & c \\ c & b \end{bmatrix}.$$

General Convexity Condition via Eigenvalues

For any symmetric Hessian $H \in \mathbb{R}^{d \times d}$,

$$H \text{ is positive definite} \iff v^\top H v > 0 \text{ for all } v \neq 0.$$

Checking this directly is hard, so we use **eigenvalues**.

- For a scalar $H = \lambda$, we have $v^\top H v = \lambda v^2 > 0$ for all $v \neq 0$ if $\lambda > 0$.
- For a diagonal matrix $H = \text{diag}(\lambda_1, \dots, \lambda_d)$, we have $v^\top H v = \sum_{i=1}^d \lambda_i v_i^2 > 0$ for all $v \neq 0$ if $\lambda_i > 0$ for all i .

General Convexity Condition via Eigenvalues

Diagonalizing Symmetric Matrices (Spectral Theorem)

For general symmetric matrix H , there exists Q such that $QQ^\top = I$ and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ such that $H = Q\Lambda Q^\top$.

What we will show: Thus for all $v \neq 0$ if $\lambda_i > 0$ for all i ,

$$v^\top H v = v^\top Q \Lambda Q^\top v = (Q^\top v)^\top \Lambda (Q^\top v) = \sum_{i=1}^d \lambda_i (q_i^\top v)^2 > 0 \implies H \succ 0.$$

Linear Algebra Concepts

- Eigenvalues and eigenvectors
- Linear independence of eigenvectors
- Basis of eigenvectors
- Diagonalizing symmetric matrices
- Spectral theorem
- Positive definiteness

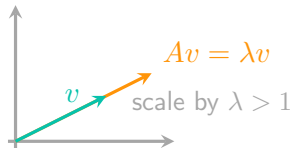
Eigenvalues and Eigenvectors

For a square matrix $A \in \mathbb{R}^{d \times d}$:

$$Av = \lambda v, \quad v \neq 0 \implies v \text{ is an eigenvector, } \lambda \text{ its eigenvalue.}$$

Geometric meaning:

- A scales (stretches or flips) v without changing its direction.
- The eigenvalues describe how much A stretches space along special directions.
- If $\lambda = 0$, that direction is *flattened* $\rightarrow$ indicates "rank deficiency".



Rank and eigenvalues:

$$\text{spec}(A) := \{\lambda \mid \exists v \neq 0, Av = \lambda v\}$$

$$\text{rank}(A) = |\{\lambda \mid \lambda \neq 0, \lambda \in \text{spec}(A)\}|$$

Takeaway: Eigenvalues reveal directions of zero stretch (null space) and quantify how much A scales space.

Independence of Eigenvectors and Basis Formation

Linear Independence

Vectors $v_1, \dots, v_n$ are **linearly independent** if

$$\sum_{i=1}^n \alpha_i v_i = \alpha_1 v_1 + \dots + \alpha_n v_n = 0 \quad \implies \quad \alpha_1 = \dots = \alpha_n = 0.$$

Otherwise, they are **linearly dependent**.

Examples: Checking Linear Independence

Example 1.

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Check if $\{v_1, v_2, v_3\}$ are linearly independent in $\mathbb{R}^2$.

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = 0 \implies (\alpha_1 + \alpha_3, \alpha_2 + \alpha_3) = (0, 0) \implies \alpha_1 = -\alpha_3, \alpha_2 = -\alpha_3.$$

$\implies$ Nontrivial solution $\implies$ **dependent**.

Geometric intuition: $v_3 = v_1 + v_2$ so it lies in the same plane, no new direction.

Examples: Checking Linear Independence

Example 2. For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, the eigenvectors $v_1 = [1, 1]^\top$, $v_2 = [1, -1]^\top$ satisfy

$$\alpha_1 v_1 + \alpha_2 v_2 = 0 \implies \alpha_1 = \alpha_2 = 0.$$

$\implies$ **independent** $\implies$ form a basis of $\mathbb{R}^2$.

Independence of Eigenvectors and Basis Formation

Theorem

If $Av_i = \lambda_i v_i$ and $Av_j = \lambda_j v_j$ with $\lambda_i \neq \lambda_j$, then v_i and v_j are linearly independent.

Proof sketch: Suppose $\alpha_i v_i + \alpha_j v_j = 0$. We claim $\alpha_i = \alpha_j = 0$. Apply A so that

$$A(\alpha_i v_i + \alpha_j v_j) = \alpha_i Av_i + \alpha_j Av_j = \alpha_i \lambda_i v_i + \alpha_j \lambda_j v_j = 0. \quad (*)$$

Additionally,

$$\lambda_j(\alpha_i v_i + \alpha_j v_j) = 0 \quad (**)$$

Taking the difference between $(*)$ and $(**)$ gives

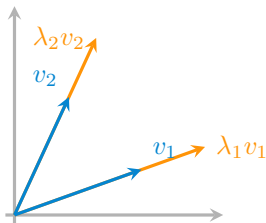
$$0 = \alpha_i(\lambda_i - \lambda_j)v_i = 0 \implies \alpha_i = 0.$$

An analogous argument holds showing $\alpha_j = 0$. Hence v_i, v_j are independent.

Independence of Eigenvectors and Basis Formation

Facts

- For $A \in \mathbb{R}^{d \times d}$ with d distinct eigenvalues, its eigenvectors $v_1, \dots, v_d$ are linearly independent.
- If A has d distinct eigenvalues, its eigenvectors form a **basis** for $\mathbb{R}^d$.
- Distinct eigenvalues guarantee independent eigenvectors, which form a basis that **diagonalizes** A .



What is a Basis?

Definition of Basis

A *basis* of a vector space $V \subseteq \mathbb{R}^d$ is a set of vectors $\{v_1, v_2, \dots, v_d\}$ such that:

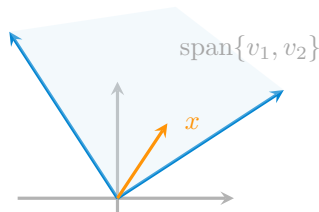
(1) $v_1, \dots, v_d$ are linearly independent, (2) they span V .

(2) $\iff$ every vector $x \in V$ can be written *uniquely* as $x = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_d v_d$.

Key Ideas:

- “Span” $\implies$ the basis vectors *reach* all of V .
- “Linear independence” $\implies$ no vector in the set is redundant.
- The number of basis vectors = **dimension** of V .

Takeaway: A basis is the “coordinate frame”: provides a unique representation of every vector.



Independence of Eigenvectors and Basis Formation

Why this matters:

- If $A \in \mathbb{R}^{d \times d}$ has d linearly independent eigenvectors, they form a **basis** for $\mathbb{R}^d$.
- In that basis, A acts simply by scaling each basis vector:

$$Av_i = \lambda_i v_i.$$

- This lets us **diagonalize** A —i.e., write it as a matrix similar to a diagonal matrix:

$$A = V\Lambda V^{-1}, \quad \text{where } V = [v_1 \ v_2 \ \dots \ v_d], \ \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d).$$

Interpretation:

- Eigenvectors give A 's *natural coordinate system*.
- Their independence guarantees we can describe any vector as

$$x = \alpha_1 v_1 + \dots + \alpha_n v_d.$$

- A then acts by simply scaling each coordinate by λ_i .

Where are we

- We have defined linear independence, and even shown how to diagonalize a matrix if it has independent eigenvectors.
- Now we want to come back to this Hessian matrix and questions about curvature.
- To apply the eigenvalue check, we need to show that Hessians can be decomposed as $H = A^T A$ for some A .

Why Symmetric Matrices Can Be Diagonalized

Claim (Spectral Theorem)

If $H \in \mathbb{R}^{d \times d}$ is **symmetric** ($H = H^\top$), then there exists an **orthogonal matrix** Q (i.e., $Q^\top Q = I$) and a diagonal matrix Λ such that

$$H = Q\Lambda Q^\top, \quad \text{where } \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d), \quad Hq_i = \lambda_i q_i.$$

Intuition:

- Symmetry $\implies$ all eigenvalues are **real**.
- Symmetry $\implies$ eigenvectors corresponding to distinct eigenvalues are **orthogonal** (i.e., $\langle v_i, v_j \rangle = 0$ for $i \neq j$).
- The eigenvectors form an orthogonal basis for $\mathbb{R}^d$.
- Then $Q^\top Q = I$ so that $HQ = Q\Lambda \implies H = Q\Lambda Q^\top$.

Why Eigenvectors of a Symmetric Matrix Are Orthogonal (Skip)

Let $H = H^\top$ and suppose

$$Hv_i = \lambda_i v_i, \quad Hv_j = \lambda_j v_j.$$

Key property of symmetric matrices: $\langle Hx, y \rangle = \langle x, Hy \rangle \quad \forall x, y.$

Compute both sides:

$$\begin{aligned}\langle Hv_i, v_j \rangle &= \lambda_i \langle v_i, v_j \rangle, \\ \langle v_i, Hv_j \rangle &= \lambda_j \langle v_i, v_j \rangle.\end{aligned}$$

Since $H = H^\top$, these must be equal:

$$(\lambda_i - \lambda_j) \langle v_i, v_j \rangle = 0 \implies \lambda_i \neq \lambda_j \implies \langle v_i, v_j \rangle = 0.$$

Intuition: Symmetry means H “preserves inner products”: it stretches along perpendicular directions, not by twisting or skewing.

What we have shown so far...

- Optimality: $\nabla f(\theta^*) = 0$ and $H(\theta^*) \succ 0 \implies \theta^*$ is a (local) minimum.

$$H(\theta) = \nabla^2 f(\theta) = \begin{bmatrix} \frac{\partial^2 f}{\partial \theta_1^2} & \cdots & \frac{\partial^2 f}{\partial \theta_1 \partial \theta_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial \theta_d \partial \theta_1} & \cdots & \frac{\partial^2 f}{\partial \theta_d^2} \end{bmatrix} \quad (\text{Hessian})$$

$H(\theta) \succ 0 \iff v^\top H(\theta) v > 0 \quad \forall v \neq 0 \implies f$ is **strictly convex** at that point

- **Eigenvalue test:** Since H is symmetric, we have that

$$H = Q\Lambda Q^\top, \quad Q^\top Q = I, \quad \text{where } \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$$

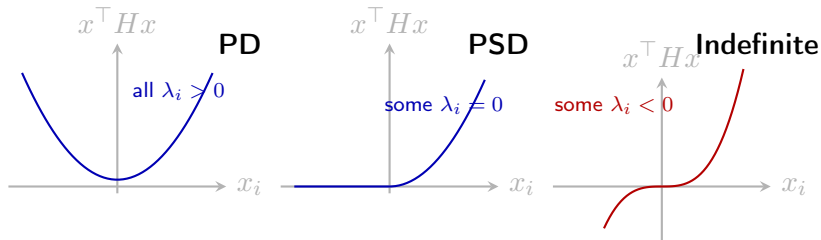
- This means that H is positive definite if and only if all its eigenvalues are positive:
 $H \succ 0 \iff$ all eigenvalues $\lambda_i > 0$.

Eigenvalues and Curvature: PD, PSD, and Indefinite

For a symmetric matrix H , curvature along direction v is measured by

$$v^\top H v = \lambda \|v\|_2^2.$$

- $\lambda_i > 0 \implies$ curves upward (bowl) • curvature in every direction $\rightarrow$ unique minimum.
- $\lambda_i = 0 \implies$ flat direction (ridge) • flat direction $\rightarrow$ multiple minima or constraints.
- $\lambda_i < 0 \implies$ curves downward (saddle) • saddle or concave directions $\rightarrow$ not convex.



Takeaway: Eigenvalues describe how a quadratic function bends in each direction. Positive eigenvalues $\implies$ upward curvature $\implies$ convex loss.

Computing Eigenvalues: The Characteristic Polynomial

Definition: For a square matrix $A \in \mathbb{R}^{n \times n}$, an **eigenvalue** λ and **eigenvector** $v \neq 0$ satisfy

$$Av = \lambda v \iff (A - \lambda I)v = 0.$$

A nontrivial solution $v \neq 0$ exists only when

$$\det(A - \lambda I) = 0.$$

This determinant is a polynomial in λ called the **characteristic polynomial** of A :

$$p_A(\lambda) = \det(A - \lambda I).$$

Why $\det(A - \lambda I) = 0$ Defines Eigenvalues

Eigenvalue equation: We seek nonzero v satisfying

$$Av = \lambda v \implies (A - \lambda I)v = 0.$$

Key fact: For any matrix M , the **homogeneous linear system** $Mv = 0$ has solutions

$$\begin{cases} v = 0 & \text{only (trivial) if } M \text{ is invertible,} \\ v \neq 0 & \text{(nontrivial) if } M \text{ is singular.} \end{cases}$$

Invertibility test: A matrix M is invertible $\iff \det(M) \neq 0$. Hence

$$(A - \lambda I)v = 0 \text{ has nonzero } v \iff \det(A - \lambda I) = 0.$$

Fact

λ is an eigenvalue of $A \iff \det(A - \lambda I) = 0$.

Why $\det(A - \lambda I) = 0$ Defines the Eigenvalues

Eigenvalue equation:

$$Av = \lambda v \quad (v \neq 0) \iff (A - \lambda I)v = 0.$$

Example.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \det(A - \lambda I) = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

$$\Rightarrow \lambda_1 = 1, \lambda_2 = 3,$$

with eigenvectors $v_1 = [1, -1]^\top$, $v_2 = [1, 1]^\top$.

Takeaway:

Eigenvalues are the roots of $\det(A - \lambda I) = 0$.

example continued

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} \implies p_A(\lambda) = \det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = (2 - \lambda)^2 - 1.$$

$$(2 - \lambda)^2 - 1 = ((2 - \lambda) - 1)((2 - \lambda) + 1) = (1 - \lambda)(3 - \lambda) = (\lambda - 1)(\lambda - 3).$$

$\Rightarrow$ eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 3$.

example continued: Eigenvectors for $\lambda = 1$

Solve $(A - \lambda I)v = 0$:

$$A - I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Row-reducing (or reading off) gives the single equation $x + y = 0$. Let $y = -x$; with parameter $t = x$ we get

$$v^{(1)} = t \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad t \neq 0.$$

Thus an eigenvector is $v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ (any nonzero scalar multiple works).

example continued: Eigenvectors for $\lambda = 3$

Solve $(A - \lambda I)v = 0$:

$$A - 3I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This yields $-x + y = 0$ (equivalently $x = y$). Let $y = x$; with parameter $t = x$ we get

$$v^{(3)} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad t \neq 0.$$

Thus an eigenvector is $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (again, any nonzero multiple).

Computing Eigenvalues: A Simple 2D Example

Matrix:

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}.$$

Step 1. Characteristic equation:

$$\det(A - \lambda I) = 0.$$

$$\det \begin{bmatrix} 3 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = (3 - \lambda)(2 - \lambda) - 1 = \lambda^2 - 5\lambda + 5 = 0.$$

Step 2. Solve for λ :

$$\lambda_{1,2} = \frac{5 \pm \sqrt{5}}{2}.$$

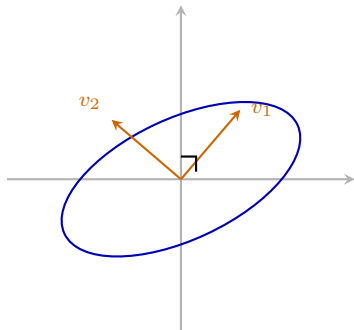
Both eigenvalues are positive $\implies A \succ 0$ (positive definite).

Computing Eigenvalues: A Simple 2D Example

Step 3. Find eigenvectors: For $\lambda_1 = \frac{5+\sqrt{5}}{2}$:

$$(A - \lambda_1 I)v_1 = 0 \implies \begin{bmatrix} 3 - \lambda_1 & 1 \\ 1 & 2 - \lambda_1 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = 0 \implies (3 - \lambda_1)v_{11} + v_{12} = 0$$

$$\implies v_{12} = (\lambda_1 - 3)v_{11} \implies v_1 \propto \begin{bmatrix} 1 \\ \lambda_1 - 3 \end{bmatrix}. \quad \text{e.g., } v_1 \approx \begin{bmatrix} 1 \\ 0.618 \end{bmatrix}, \lambda_1 = \frac{5+\sqrt{5}}{2}$$



Takeaway: Eigenvalues give the *stretch factors*, and eigenvectors the *principal directions* of the quadratic form $x^\top Ax$.

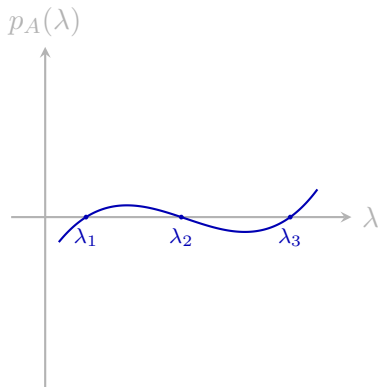
Eigenvalues in Higher Dimensions: 3×3 case

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \implies p_A(\lambda) = \begin{vmatrix} 2 - \lambda & 1 & 0 \\ 1 & 2 - \lambda & 1 \\ 0 & 1 & 2 - \lambda \end{vmatrix}.$$

Expanding the determinant:

$$p_A(\lambda) = -\lambda^3 + 6\lambda^2 - 10\lambda + 4.$$

Setting $p_A(\lambda) = 0$ gives the three eigenvalues.



3×3 Example

Matrix (symmetric, block diagonal) & Characteristic polynomial:

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}. \quad p_A(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} 2 - \lambda & 0 & 0 \\ 0 & 2 - \lambda & 1 \\ 0 & 1 & 2 - \lambda \end{bmatrix}.$$

Because the first row/column isolates,

$$p_A(\lambda) = (2 - \lambda) \det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = (2 - \lambda)((2 - \lambda)^2 - 1).$$

$$\Rightarrow \boxed{p_A(\lambda) = (2 - \lambda)(1 - \lambda)(3 - \lambda)} = -(\lambda - 1)(\lambda - 2)(\lambda - 3).$$

Eigenvalues:

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 3.$$

3×3 Example: Eigendecomposition

Eigenvectors:

- For $\lambda = 2$: any vector supported on the first coordinate, e.g. $v_2 = (1, 0, 0)^\top$.
- For $\lambda = 3$: solve $\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \implies x = y$, so pick $v_3 = (0, 1, 1)^\top$.
- For $\lambda = 1$: solve $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \implies x = -y$, so pick $v_1 = (0, 1, -1)^\top$.

Takeaway: Using a block (or nearly block) structure makes $p_A(\lambda)$ factor at sight.

Eigendecomposition Continued

Characteristic polynomial:

$$p_A(\lambda) = \det(A - \lambda I) = (2 - \lambda)((2 - \lambda)^2 - 1) = (2 - \lambda)(1 - \lambda)(3 - \lambda).$$

Eigenvalues: $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3.$

Eigenvectors (unnormalized):

$$\lambda_1 = 1 : v_1 = (0, 1, -1)^\top \quad \lambda_2 = 2 : v_2 = (1, 0, 0)^\top \quad \lambda_3 = 3 : v_3 = (0, 1, 1)^\top$$

Orthonormal basis:

$$\hat{v}_1 = \frac{1}{\sqrt{2}}(0, 1, -1)^\top, \quad \hat{v}_2 = (1, 0, 0)^\top, \quad \hat{v}_3 = \frac{1}{\sqrt{2}}(0, 1, 1)^\top.$$

Eigendecomposition Continued

Assemble Q and Λ :

$$Q = [\hat{v}_1 \ \hat{v}_2 \ \hat{v}_3] = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \Lambda = \text{diag}(1, 2, 3).$$

$$\boxed{A = Q \Lambda Q^\top} \quad (\text{orthogonal diagonalization; } Q^\top Q = I).$$

Eigenvalues in Higher Dimensions: The Characteristic Polynomial

In general:

$$p_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_1 \lambda + c_0.$$

- The coefficients depend on the entries of A .
- The **roots** $\lambda_1, \dots, \lambda_n$ are the eigenvalues.
- If A is symmetric, all λ_i are real.

Takeaway: Eigenvalues are the **roots of the characteristic polynomial** $\det(A - \lambda I) = 0$.
Each eigenvalue corresponds to a direction where A acts like simple scaling.

Calculus & Linear Algebra Summary

- **Optimality conditions:** $\nabla f(\theta^*) = 0$ and $H(\theta^*) \succ 0 \implies \theta^*$ is a (local) minimum.
- **Symmetric matrices can be diagonalized:** $H = Q\Lambda Q^\top$ for some orthogonal matrix Q and diagonal matrix Λ with eigenvalues $\lambda_1, \dots, \lambda_d$.
- **Positive definiteness:** $H \succ 0 \iff$ all eigenvalues $\lambda_i > 0$.
- **Computing eigenvalues:** Eigenvalues are the roots of the characteristic polynomial $\det(A - \lambda I) = 0$.

Solving Least Squares (Coordinate Form)

For one data point: $f_{\theta}(x_i) = \theta_1 x_{i1} + \theta_2 x_{i2} + \cdots + \theta_d x_{id} = \theta^{\top} x$.

$$\text{minimize } \mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \theta_1 x_{i1} - \theta_2 x_{i2} - \cdots - \theta_d x_{id})^2.$$

Take partial derivatives with respect to each θ_j :

$$\frac{\partial \mathcal{L}}{\partial \theta_j} = -\frac{2}{n} \sum_{i=1}^n x_{ij} \left(y_i - \sum_{k=1}^d \theta_k x_{ik} \right).$$

normal equations: $\frac{\partial \mathcal{L}}{\partial \theta_j} = 0 \iff \sum_{i=1}^n x_{ij} y_i = \sum_{i=1}^n \left(x_{ij} \left(\sum_{k=1}^d x_{ik} \theta_k \right) \right), \quad j = 1, \dots, d.$

These can be solved as a system of d linear equations in θ .

Takeaway: Find the min by setting each derivative to zero: i.e., first-order optimality.

Gradients of Linear and Quadratic Forms

Linear Forms

Let $x, b \in \mathbb{R}^n$. Then

$$\nabla_x(b^\top x) = b, \quad \nabla_x(x^\top b) = b.$$

Quadratic Form

Let $A \in \mathbb{R}^{n \times n}$ and define

$$f(x) = x^\top Ax.$$

Then

$$\nabla_x(x^\top Ax) = (A + A^\top)x.$$

If A is symmetric, this simplifies to

$$\nabla_x(x^\top Ax) = 2Ax.$$

Gradients and Quadratic Forms

General Quadratic Function

For

$$f(x) = \frac{1}{2}x^\top Qx + b^\top x + c, \quad Q = Q^\top,$$

we have

$$\nabla_x f(x) = Qx + b.$$

Example: Least Squares

$$f(x) = \|Ax - y\|_2^2 \quad \Rightarrow \quad \nabla_x f(x) = 2A^\top(Ax - y).$$

If $f(x) = \frac{1}{2}\|Ax - y\|_2^2$, then

$$\nabla_x f(x) = A^\top(Ax - y).$$

Solving Least Squares (Vector Form)

Recall the objective:

$$\mathcal{L}(\theta) = \frac{1}{n} \|y - X\theta\|_2^2 = \frac{1}{n} (y - X\theta)^\top (y - X\theta).$$

Compute the gradient: $\nabla_{\theta} \mathcal{L}(\theta) = -\frac{2}{n} X^\top (y - X\theta).$

optimality conditions: $\nabla_{\theta} \mathcal{L}(\theta) = 0 \implies X^\top X \hat{\theta} = X^\top y.$

If $X^\top X$ is **invertible** (columns of X are independent),

$$\boxed{\hat{\theta} = (X^\top X)^{-1} X^\top y.}$$

- Gradient = 0 $\implies$ flat point (no descent direction).
- Since $\mathcal{L}$ is **convex** (quadratic, positive-definite Hessian), this is the global minimum.

Coordinate vs. Vector Gradient (2D Example)

Consider two features: $x_i = [x_{i1}, x_{i2}]^\top$ and parameters $\theta = [\theta_1, \theta_2]^\top$.

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - x_{i1}\theta_1 - x_{i2}\theta_2)^2 = \frac{1}{n} \|y - X\theta\|_2^2.$$

Coordinate form:

$$\frac{\partial \mathcal{L}}{\partial \theta_1} = -\frac{2}{n} \sum_{i=1}^n x_{i1}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2),$$

$$\frac{\partial \mathcal{L}}{\partial \theta_2} = -\frac{2}{n} \sum_{i=1}^n x_{i2}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2).$$

Vector form:

$$\begin{aligned} \nabla_{\theta} \mathcal{L}(\theta) &= -\frac{2}{n} X^\top (y - X\theta) \\ &= -\frac{2}{n} \begin{bmatrix} \sum_i x_{i1}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2) \\ \sum_i x_{i2}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2) \end{bmatrix}. \end{aligned}$$

Convexity of the Least Squares Objective

$$\mathcal{L}(\theta) = \frac{1}{n} \|y - X\theta\|_2^2 = \frac{1}{n} (y - X\theta)^\top (y - X\theta).$$

Gradient and Hessian:

$$\nabla_{\theta} \mathcal{L}(\theta) = -\frac{2}{n} X^\top (y - X\theta),$$

$$\nabla_{\theta}^2 \mathcal{L}(\theta) = \frac{2}{n} X^\top X.$$

Convexity check:

$$v^\top \nabla_{\theta}^2 \mathcal{L}(\theta) v = \frac{2}{n} \|Xv\|_2^2 \geq 0.$$

When is it *strictly* convex?

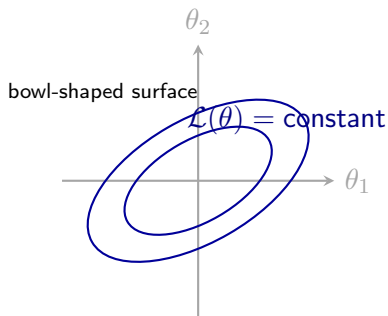
If the columns of X are linearly independent ($X^\top X$ invertible), then

$$v^\top X^\top X v = \|Xv\|_2^2 > 0 \quad \text{for all } v \neq 0.$$

So $\nabla^2 \mathcal{L}(\theta) \succ 0$ (**positive definite**).

observe: $X^\top X$ is symmetric and PSD!

Convexity Intuition (2D Example)



Takeaway: Least squares is convex because its Hessian $\frac{2}{n}X^\top X$ is positive definite when X has full column rank.

What happens if some directions have zero curvature?

Back to the optimization problem

- Any vector $\hat{\theta}$ satisfying the following is a solution (i.e., a least squares approximate solution):

$$\|X\hat{\theta} - y\|_2^2 \leq \|X\theta - y\|_2^2 \quad \forall \theta \in \mathbb{R}^d$$

- It need not be the case that $X\hat{\theta} = y$!
- Regression:** We say that $\hat{\theta}$ is the result of *regressing* the vector y onto the columns of X .

Column Interpretation

$$X = \begin{bmatrix} | & \cdots & | \\ x_1 & \cdots & x_d \\ | & \cdots & | \end{bmatrix}, \quad x_i \in \mathbb{R}^n$$

- Least squares problem is equivalent to finding a linear combination of the columns that is closest to $y \in \mathbb{R}^n$:

$$\|X\theta - y\|_2^2 = \|\theta_1 x_1 + \cdots + \theta_d x_d - y\|_2^2$$

- For a solution $\hat{\theta}$, we have that $X\hat{\theta} = \hat{\theta}_1 x_1 + \cdots + \hat{\theta}_d x_d$
- $X\hat{\theta}$ is the *closest* (in Euclidean distance) to $y \in \mathbb{R}^n$ among all linear combinations of vectors $x_1, \dots, x_d \in \mathbb{R}^n$

Row Interpretation

$$X = \begin{bmatrix} \text{---} & \tilde{x}_1^\top & \text{---} \\ \vdots & \dots & \vdots \\ \text{---} & \tilde{x}_n^\top & \text{---} \end{bmatrix}, \quad \tilde{x}_i \in \mathbb{R}^d$$

- Recall that $r = X\theta - y$ is the residual vector
- entries of r are then given by $r_i = \tilde{x}_i^\top \theta - y_i, \quad i = 1, \dots, n$
- objective can be rewritten as

$$\|X\theta - y\|_2^2 = (\tilde{x}_1^\top \theta - y_1)^2 + \dots + (\tilde{x}_n^\top \theta - y_n)^2$$

◆ Row Space (or “geometry in parameter space”) — the row picture

Interpretation:

Each row of X represents an observation:

$$x_i^\top \theta \approx y_i.$$

So each row defines a **hyperplane** in $\mathbb{R}^n$:

$$\mathcal{H}_i = \{\theta \in \mathbb{R}^n \mid x_i^\top \theta = y_i\}.$$

The least-squares solution $\hat{\theta}$ is the point in $\mathbb{R}^n$ that **comes closest (in total squared distance)** to all these hyperplanes.

Geometric view:

- If the system $X\theta = y$ is consistent, all hyperplanes intersect at a common point (the exact solution).
- If it's **overdetermined**, they don't intersect — least squares finds the point $\hat{\theta}$ that minimizes the perpendicular distances to all hyperplanes.
- The set of feasible directions that don't change $X\theta$ is the **null space** $\mathcal{N}(X)$.

Example

$$2c_1=0, -c_1+c_2=0, 2c_2=0$$

$$X = \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad n = 3, \quad d = 2$$

- $X\theta = y \implies \{2\theta_1 = 1, -\theta_1 + \theta_2 = 0, 2\theta_2 = -1\}$ a system that has no solution
- Least squares problem: using row interpretation we have

$$\min_{\theta_1, \theta_2} \{(2\theta_1 - 1)^2 + (-\theta_1 + \theta_2)^2 + (2\theta_2 + 1)^2\}$$

Recall: Finding Minima via Calculus

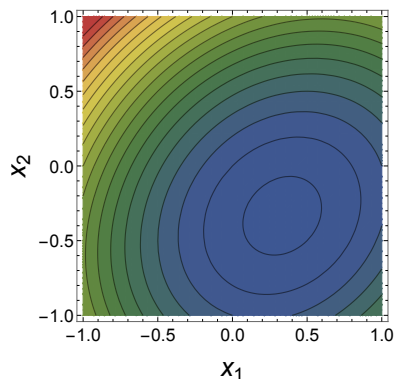
[VMLS, App. C]

- **Calculus:** to find $\min_x f(x)$, we set $\frac{d}{dx} f(x) = 0$ and find x^* that solves the equation, and check that $\frac{d^2}{dx^2} f(x)|_{x=x^*} > 0$
- **Multivariable Case:**
 - ▶ $\nabla f(x) = 0 \iff \left\{ \frac{\partial}{\partial x_i} f(x) = 0, i = 1, \dots, n \right\}$
 - ▶ Hessian: $\nabla^2 f(x)|_{x=x^*} \succ 0$ where

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} \quad \text{and} \quad \nabla^2 f(x)|_{x=x^*} \succ 0 \iff \text{eigenvalues positive}$$

Example Continued

$$\min_{\theta_1, \theta_2} \{ (2\theta_1 - 1)^2 + (-\theta_1 + \theta_2)^2 + (2\theta_2 + 1)^2 \}$$



- setting the derivative to zero and solving for θ_1 and θ_2 we get $\theta_1 = \frac{1}{3}$ and $\theta_2 = -\frac{1}{3}$
- Hence, the solution is $\theta = \begin{bmatrix} \frac{1}{3} \\ -\frac{1}{3} \end{bmatrix}$
- Check rank of X : Does $c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2 = 0$ imply $c_1 = c_2 = 0$? Yes!

Least Squares Solution via Calculus

Assumption [A1]: The columns of X are linearly independent; i.e.,

$$\sum_{i=1}^n c_i x_i = 0 \iff c_i = 0 \quad \forall i = 1, \dots, n$$

- Any minimizer $\hat{x}$ of $f(\theta) = \|X\theta - y\|_2^2$ must satisfy

$$\frac{\partial f}{\partial \theta_i}(\hat{\theta}) = 0, \quad i = 1, \dots, n \iff \nabla f(\hat{\theta}) = 0$$

- In matrix form, the gradient is

$$\nabla f(\theta) = 2X^\top(X\theta - y) \quad [\text{VMLS, p. 184}]$$

Least Squares Solution via Calculus Continued

- Any minimizer $\hat{\theta}$ of $f(\theta) = \|X\theta - y\|_2^2$ must satisfy

$$\nabla f(\hat{\theta}) = 2X^\top(X\hat{\theta} - y) = 0 \iff X^\top X\hat{\theta} = X^\top y \quad \text{[normal equations]}$$

- Gram matrix:** entries of $X^\top X$ are inner products of the columns of X
- [A1]** $\implies X^\top X$ is invertible [VMLS, §11.5, pg. 214]: Consider $\theta \in \mathbb{R}^d$ such that $X^\top X\theta = 0$. multiplying on the left by $\theta^\top$ we get

$$0 = \theta^\top 0 = \theta^\top X^\top X\theta = \|X\theta\|_2^2 \implies X\theta = 0.$$

Since the only solution to $X\theta = 0$ is $\theta = 0$, we have that $X^\top X$ is invertible.

- Hence, $\hat{\theta} = (X^\top X)^{-1}X^\top y$ is the *only* solution of the normal equations
- Pseudo-inverse:**
- $\hat{\theta} = X^\dagger y$ solves $X\theta = y$ **if** the set of equations has a solution, otherwise it's called the **least squares approximate solution**.

Direct Verification of the Solution

- Let's check via direct verification: show that for any $\theta \neq \hat{\theta} = X^\dagger y$ we have

$$\|X\hat{\theta} - y\|_2^2 < \|X\theta - y\|_2^2$$

- Indeed,

$$\begin{aligned}\|X\theta - y\|_2^2 &= \|(X\theta - X\hat{\theta}) + (X\hat{\theta} - y)\|_2^2 \\ &= \|X(\theta - \hat{\theta})\|_2^2 + \|X\hat{\theta} - y\|_2^2 + 2(\theta - \hat{\theta})^\top X^\top (X\hat{\theta} - y)\end{aligned}$$

since $\|u + v\|_2^2 = (u + v)^\top (u + v) = \|u\|_2^2 + \|v\|_2^2 + 2u^\top v$

- Claim:** $(\theta - \hat{\theta})^\top X^\top (X\hat{\theta} - y) = 0$

proof: since $(X^\top X)\hat{\theta} = X^\top y$,

$$(\theta - \hat{\theta})^\top X^\top (X\hat{\theta} - y) = (\theta - \hat{\theta})^\top (X^\top X\hat{\theta} - X^\top y) = 0$$

Why "Pseudo-Inverse"?

For a square invertible matrix A , the inverse A^{-1} satisfies $A^{-1}A = I$.

Why "Pseudo-Inverse"?

For a non-square matrix $X \in \mathbb{R}^{n \times d}$ with $n > d$ (tall, full column rank), there's no true inverse. But $(X^T X)^{-1} X^T$ acts like a "left inverse":

$$(X^T X)^{-1} X^T \cdot X = (X^T X)^{-1} (X^T X) = I_d$$

It inverts X "as much as possible." This is the Moore-Penrose pseudoinverse X^+ .

Key property: If $X\theta = \mathbf{y}$ has no exact solution (overdetermined system), then $\theta = X^+ \mathbf{y}$ gives the least squares solution minimizing $\|X\theta - \mathbf{y}\|^2$.

The Pseudo-Inverse is a Projection Matrix

The Pseudo-Inverse is a Projection Matrix

The matrix $P = X(X^\top X)^{-1}X^\top$ is the orthogonal projection onto the column space of X .

Properties:

- $P^2 = P$ (idempotent): Projecting twice = projecting once
- $P^\top = P$ (symmetric): Orthogonal projection
- $P\mathbf{y}$ is the closest point in $\text{col}(X)$ to $\mathbf{y}$
- $(I - P)$ projects onto the orthogonal complement (residuals in regression)

In linear regression $\hat{\mathbf{y}} = X\hat{\boldsymbol{\theta}} = X(X^\top X)^{-1}X^\top \mathbf{y} = P\mathbf{y}$ is the projection of $\mathbf{y}$ onto the column space.

Direct Verification of the Solution

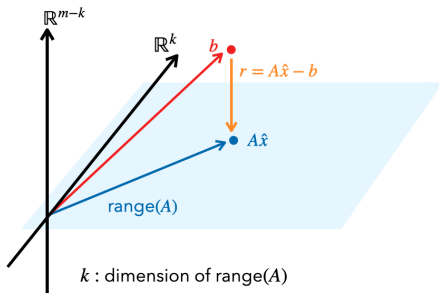
- Coming back to the expression for the objective,

$$\|X\theta - y\|_2^2 = \underbrace{\|X(\theta - \hat{\theta})\|_2^2}_{\geq 0} + \|X\hat{\theta} - y\|_2^2$$

- Hence, $\|X\hat{\theta} - y\|_2^2 \leq \|X\theta - y\|_2^2$
- **Row form of solution:** sometimes its useful to express the solution as

$$\hat{\theta} = X^\dagger y = (X^\top X)^{-1} X^\top y = \left(\sum_{i=1}^m \tilde{x}_i \tilde{x}_i^\top \right)^{-1} \left(\sum_{i=1}^m y_i \tilde{x}_i \right)$$

Orthogonality Principle

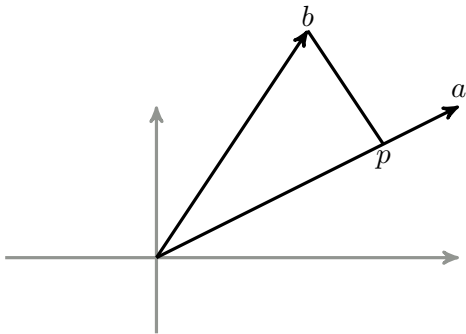


- $A\hat{x}$ is the linear combination of columns of A closest to b
- Residual $r = A\hat{x} - b$ satisfies the **orthogonality principle**:

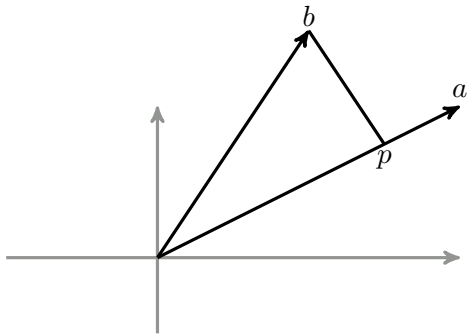
$$(Az) \perp r \quad \forall z \in \mathbb{R}^n$$

- **Why?**

Let's Look at the Vector Case



Let's Look at the Vector Case



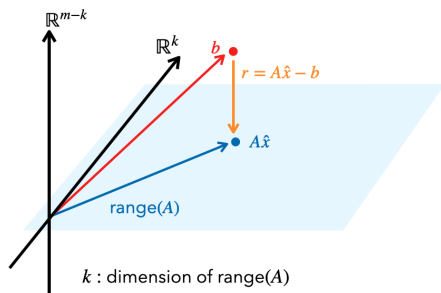
- Since p lies along the vector a , we know that $p = xa$ for some x
- Also, a is perpendicular to $r = b - xa$:

$$a^\top(b - xa) = 0 \implies xa^\top a = a^\top b$$

$$x = \frac{a^\top b}{a^\top a} \text{ and } p = ax = a \frac{a^\top b}{a^\top a}$$

- projection matrix: $P = a(a^\top a)^{-1}a^\top$

Orthogonality Principle



- $X\hat{\theta}$ is the linear combination of columns of X closest to y
- Residual $r = X\hat{\theta} - y$ satisfies the so **orthogonality principle**:

$$(Xz) \perp r \quad \forall z \in \mathbb{R}^n$$

- **Why?**

- First, **[normal equations]** $\iff X^\top(X\hat{\theta} - y) = 0$
- Hence, for any $z \in \mathbb{R}^n$, we have

$$(Xz)^\top r = (Xz)^\top (X\hat{\theta} - y) = z^\top X^\top (X\hat{\theta} - y) = 0$$

Projection

- The least squares solution is $\hat{\theta} = (X^\top X)^{-1} X^\top y$ and the prediction is $\hat{y} = X\hat{\theta}$
- $P = X(X^\top X)^{-1} X^\top$ is a projection matrix: it projects onto the subspace formed by the columns of X
 - ▶ P is a projection matrix if $P^2 = P$
- Orthogonal decomposition:

$$y = y_{\mathcal{R}(X)} + y_{\mathcal{R}(X)^\perp} \quad \text{where} \quad y_{\mathcal{R}(X)} = X\hat{\theta} \quad \text{and} \quad y_{\mathcal{R}(X)^\perp} = r = X\hat{\theta} - y$$

More Examples

Consider

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$$

Find the least squares approximate solution to $Ax = b$.

Solution. First

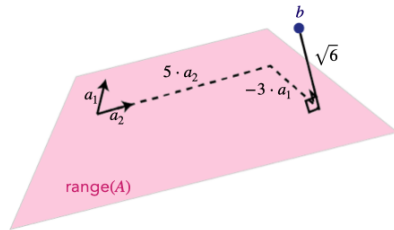
$$A^T A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 3 & 3 \end{bmatrix} \quad \text{and} \quad A^T b = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

$$(A^T A)^{-1} = \frac{1}{15 - 9} \begin{bmatrix} 3 & -3 \\ -3 & 5 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{5}{6} \end{bmatrix} \implies \hat{x} = \underbrace{\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{5}{6} \end{bmatrix}}_{(A^T A)^{-1}} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

Example Continued

- The solution minimizes the distance from $A\hat{x}$ to b :

$$\|b - A\hat{x}\|_2^2 = \left\| \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix} \right\|_2^2 = \left\| \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\|_2^2$$



What about under-determined systems?

Minimum Norm Solution and the Null Space

One particular solution to the linear system $Ax = y$ is

$$x_{\text{ln}} = A^{\top}(AA^{\top})^{-1}y,$$

where $AA^{\top}$ is invertible since A has full rank.

In fact, x_{ln} is the solution of $Ax = y$ that minimizes the norm $\|x\|$; that is,

$$x_{\text{ln}} = \arg \min_{x \in \mathbb{R}^n} \|x\| \quad \text{subject to } Ax = y.$$

Proof of Minimal Norm Property.

Suppose $Ax = y$. Then

$$A(x - x_{\text{ln}}) = 0,$$

and

$$\begin{aligned}(x - x_{\text{ln}})^{\top} x_{\text{ln}} &= (x - x_{\text{ln}})^{\top} A^{\top} (AA^{\top})^{-1} y \\ &= (A(x - x_{\text{ln}}))^{\top} (AA^{\top})^{-1} y \\ &= 0.\end{aligned}$$

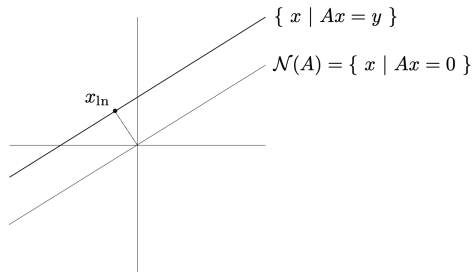
Hence $(x - x_{\text{ln}}) \perp x_{\text{ln}}$.

Then,

$$\|x\|^2 = \|x_{\text{ln}} + (x - x_{\text{ln}})\|^2 = \|x_{\text{ln}}\|^2 + \|x - x_{\text{ln}}\|^2 \geq \|x_{\text{ln}}\|^2.$$

Thus, x_{ln} has the smallest norm of any solution to $Ax = y$.

Null Space Connection



- orthogonality condition: $x_{\text{In}} \perp \text{null}(A)$
- projection interpretation: x_{In} is the projection of 0 onto the solution set $\{x \mid Ax = y\}$

Pseudo-Inverse and Projection Formulas

Full Rank **Fat** Matrix A

- $A^\dagger = A^\top (AA^\top)^{-1}$ is called the *pseudo-inverse* of full rank, fat A .
- $A^\top (AA^\top)^{-1}$ is a *right inverse* of A .
- $I - A^\top (AA^\top)^{-1}A$ gives projection onto $\mathcal{N}(A)$.

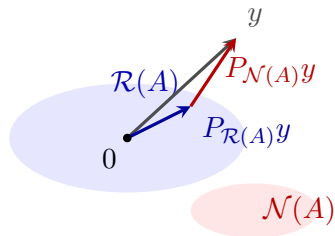
Analogous Formulas for Full Rank **Skinny** Matrix A

- $A^\dagger = (A^\top A)^{-1}A^\top$
- $(A^\top A)^{-1}A^\top$ is a *left inverse* of A .
- $A(A^\top A)^{-1}A^\top$ gives projection onto $\mathcal{R}(A)$.

Geometry of Pseudo-Inverse and Projections

Projection Interpretation

- $P_{\mathcal{R}(A)} = A(A^\top A)^{-1}A^\top$ projects onto the **range** of A .
- $P_{\mathcal{N}(A)} = I - A^\top(AA^\top)^{-1}A$ projects onto the **null space** of A .
- $A^\dagger$ maps y to the minimum-norm x such that $Ax = y$.



$$y = P_{\mathcal{R}(A)}y + P_{\mathcal{N}(A)}y, \quad \text{with } P_{\mathcal{R}(A)}P_{\mathcal{N}(A)} = 0.$$

Next time

Today we covered:

- Convexity, Hessians, and curvature
- Linear independence, basis, and dimension
- Why symmetric matrices can be diagonalized
- Checking convexity with the eigenvalue test
- Solving least squares by setting the gradient to zero

Next time we will cover:

- Linear Algebra Concepts: Matrix Rank and the Pseudoinverse, Under- vs Over-determined systems
- Projections onto subspaces and connections to least squares
- Meaning/interpretation of least squares as a ML model